

SGDrive: Scene-to-Goal Hierarchical World Cognition for Autonomous Driving

Jingyu Li^{1,2*} Junjie Wu^{3*} Dongnan Hu^{4,2} Xiangkai Huang³ Bin Sun³ Zhihui Hao³ Xianpeng Lang³ Xiatian Zhu⁵ Li Zhang^{1,2}

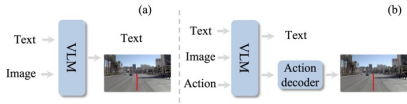
¹ Fudan University ² Shanghai Innovation Institute ³ Li Auto Inc. ⁴ Tongji University ⁵ University of Surrey

<https://github.com/LogosRoboticsGroup/SGDrive>



Poster

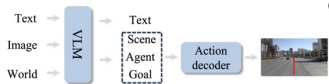
Motivation



Previous methods suffer from several key limitations:

- Lack of spatial perception
- Difficulty in discerning critical information
- Lack of future world state forecast

Key points and contributions



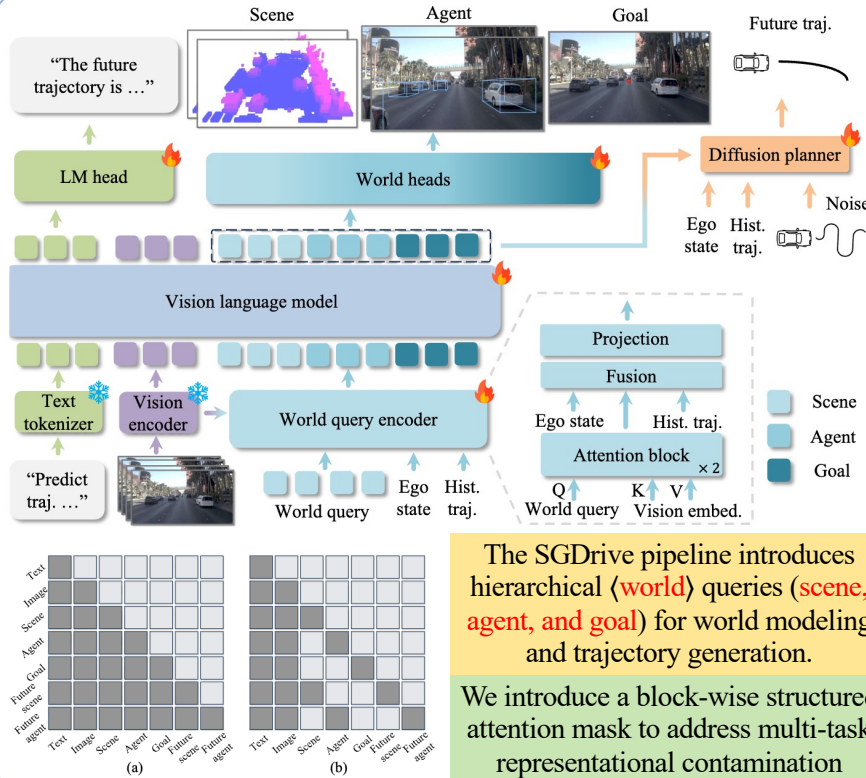
Our SGDrive acquires comprehensive driving-relevant world knowledge from three aspects:

- Scene geometric layout
- Perceiving driving-relevant agents
- Inferring driving goal

Contributions:

- We propose a novel Vision-Language-Action model with world representation for autonomous driving.
- We unlock the world cognition capability of VLA models.
- SGDrive achieves state-of-the-art performance on the NAVSIM benchmark

SGDrive overview



The SGDrive pipeline introduces hierarchical **<world>** queries (**scene, agent, and goal**) for world modeling and trajectory generation.

We introduce a block-wise structured attention mask to address multi-task representational contamination

Experiments

Table 1. Performance comparison on NAVSIM v1 *navtest* using closed-loop metrics. We additionally report results with reinforcement learning fine-tuning (RFT) to enable fair comparison with methods that adopt such training strategies. [†] denotes models fine-tuned on the NAVSIM trajectory dataset. Bold indicates the best results under SFT and RFT settings, respectively.

Method	Image	Lidar	NC [†]	DAC [†]	TTC [†]	Conf. [†]	EP [†]	PDMS [†]
VADv2-V ₁₀₂ [6]	✓	✓	97.2	89.1	91.6	100	76.0	80.9
UniAD [14]	✓	✓	97.8	91.9	92.9	100	78.8	83.4
TransFuser [43]	✓	✓	97.7	92.8	92.8	100	79.2	84.0
PARA-Drive [58]	✓	✓	97.9	92.4	93.0	99.8	79.3	84.0
DRAMA [64]	✓	✓	98.0	93.1	94.8	100	80.1	85.5
Epona [70]	✓	✓	97.9	95.1	93.8	99.9	80.4	86.2
Hydra-MDP-V ₁₀₂ -W-EP [32]	✓	✓	98.3	96.0	94.6	100	78.7	86.5
ARTEMIS [11]	✓	✓	98.3	95.1	94.3	100	81.4	87.0
DiffusionDrive [36]	✓	✓	98.2	96.2	94.7	100	82.2	88.1
WoTE [29]	✓	✓	98.5	96.8	94.9	99.9	81.9	88.3
VLMs-based Methods (SFT)								
AutoVLA-3B [75]	✓	✓	96.9	92.4	88.1	99.1	75.8	80.5
QwenVL2.5-8B [†] [2]	✓	✓	97.8	92.1	92.8	100	78.3	83.3
InternVL3-8B [†] [77]	✓	✓	97.0	92.4	91.8	100	78.9	83.3
ImagiDrive-A-7B [27]	✓	✓	97.7	95.3	93.0	99.9	80.4	86.0
ReCogDrive-2B [30]	✓	✓	98.1	94.7	94.2	100	80.9	86.5
ReCogDrive-8B [30]	✓	✓	98.3	95.1	94.3	100	81.1	86.8
SGDrive-2B (ours)	✓	✓	98.6	95.1	95.4	100	81.2	87.4
VLMs-based Methods (RFT)								
AutoVLA-3B [75]	✓	✓	98.4	95.6	98.0	99.9	81.9	89.1
ReCogDrive-2B [30]	✓	✓	97.9	97.3	94.9	100	87.3	90.8
ReCogDrive-8B [30]	✓	✓	97.8	97.7	94.9	100	86.3	90.5
SGDrive-2B (ours)	✓	✓	98.6	97.8	96.2	100	85.8	91.1

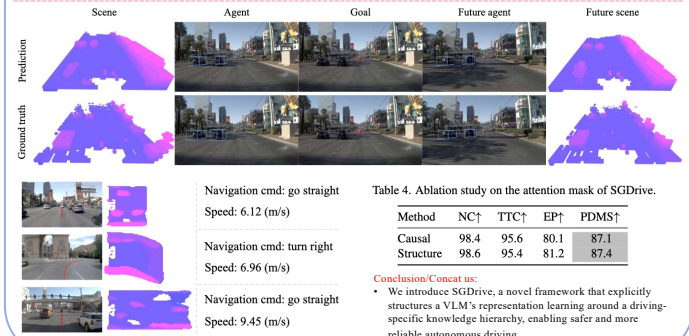


Table 4. Ablation study on the attention mask of SGDrive.

Method	NC [†]	TTC [†]	EP [†]	PDMS [†]
Causal	98.4	95.6	80.1	87.1
Structure	98.6	95.4	81.2	87.4

Conclusion/Concat us:

- We introduce SGDrive, a novel framework that explicitly structures a VLM’s representation learning around a driving-specific knowledge hierarchy, enabling safer and more reliable autonomous driving.
- Email: jingyuli0911@gmail.com